MIL-I-13504E
24 November 1969
SUPERSEDING
MIL-I-0013504D(MU)
9 February 1966
MIL-I-13504B
28 January 1961

MILITARY SPECIFICATION

IGNITER, BOMB, WP, AN-M23A1 (W/O Fuze)

This specification is mandatory for use by all Departments and Agenices of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of igniter filled with white phosphorus (WP).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel Strapping, Flat.
TT-W-572	-	Wood Preservative, Water Repellant.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	-	Paperboard, Wrapping, Cushioning.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-2427	-	Boxes, Ammunition Packing, Wood, Nailed.
MIL-S-8660	-	Silicone Compound.
MIL-C-21567	-	Compound, Silicone, Soft Film.

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STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL14-5-2151 - Igniter, Bomb, WP, AN-M23A1 (w/o Fuze).
IDL14-5-2151 - Igniter, Bomb, WP, AN-M23A1 (w/o Fuze) Inspection Data List.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specifiction to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL14-5-2151 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the igniter shall conform to the specification and drawings cited on Drawing DL14-5-2151 and subsidiary drawings.

3.2 Assembly. The igniter shall be assembled as shown on Drawing C14-5-2151 (see 6.3).

3.3 Filling.

3.3.1 Level. When examined in accordance with 4.2.4.1, a 17.3 to 25 percent void, of total volume shall exist above the surface of the water over the WP in the filled and closed igniter. The igniter shall be in the filling-end-up position and the WP at a temperature of 140° ± 10° F.

3.3.2 Procedure. The igniter body assembly shall be thoroughly cleaned, free from foreign matter and filled with molten WP. If the igniter body is filled under water, the depth of water remaining on the surface of the WP shall not exceed 1/8 inch (see 6.5) when tested as specified in 4.2.4.1. The igniter shall be in a vertical position, filling-end-up, when the molten WP solidifies.

3.4 Cleaning. When tested in accordance with 4.2.4.2, the exterior surface of the filled and closed igniter and the interior of the burster well shall be thoroughly cleaned to remove any grease, oil, WP or other contamination (see 6.6).

3.5 Leakage.

3.5.1 Igniter. The closed and cleaned igniter shall not leak when maintained at a temperature of 207° to 212° F for a minimum period of 15 minutes and tested as specified in 4.2.4.3(a). Alternate method: The igniter shall not leak when maintained at a temperature of 160° ± 5° F for a minimum period of 8 hours and tested as specified in 4.2.4.3(a).

3.5.2 Container. The sealed container shall not leak when submerged to a maximum depth of 2 inches below the surface of the water at a temperature of 185° ± 4° F for 15 seconds when tested as specified in 4.2.4.3(b).

3.6 Workmanship. The igniter shall be free from foreign matter (3.4) and damage such as cracks, dents or deformation that could affect its intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herien, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisified.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the igniters of one interfix number, produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.2.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination. The sample igniters and the level A preparation for delivery shall be examined in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each item in the lot shall be tested for the critical characteristics listed in the classification of defects.

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4.2.3.3 Classification of defects. (See inspection data list IDL 14-5-2151 for inspection equipment required to perform the examinations specified in this section.)

(a) Igniter, bomb, WP, AN-M23A1 (w/o fuze) (Drawing C14-5-2151).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.2.4.3(a)
2	WP contamination	4.2.4.2
Major: AQL 1.0 percent defective		
101	Filling excessive	
102	Filling insufficent	
103	Excessive water in igniter	4.2.4.1
104	Marking missing or incorrect	
105	Threads not coated	
106	Protective finish missing or incorrect	
107	Plug incorrect (0.031 above flush - 0.156 below flush)	
108	Workmanship (3.6)	

(b) Preparation for delivery (section 5).

Critical:	None defined	
Major: AQL 0.25 percent defective		
101	Container leakage	4.2.4.3(b)
AQL 2.5 percent defective		
102	Packaging incorrect	
103	Loose pack	
104	Container closure incorrect	
105	Container damaged (broken or wracked)	
106	Strapping loose or missing	
107	Marking missing or incorrect	

4.2.4 Tests. Tests shall be conducted as follows: (See inspection data list IDL 14-5-2151 for inspection equipment required to perform the tests specified in this section.)

4.2.4.1 Level. With the WP in a molten condition, the filling height and depth of water remaining on the surface of the WP in the filled igniter shall be radiographed with the igniter in the filling-end-up position.

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4.2.4.2 Cleaning. The cleaning procedure and the method for determining WP contamination shall be approved by the Government inspector (see 6.6).

4.2.4.3 Leakage.

(a) Igniter. The filled and closed igniter shall be placed in an oven in a vertical, filling-end-up position for the 207° to 212° F method. For the alternate 160 ° + 5° F method, the igniter shall be placed in a filling-end-down position. The igniter shall be maintained at the specified temperature for the required period of time and observed for leakage. When tested at 207° to 212° F, the munition temperature shall be measured by thermocouples or thermometers inserted in the burster wells of at least two munition located in each oven batch. Leakage shall be determined by visual examination.

(b) Container. The sealed container shall be submerged in water for the required period of time at the specified depth and water temperature. The emission of air bubbles shall constitute leakage.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and preservation. Unpainted surfaces of the filled and closed igniter, including threaded surfaces, shall be cleaned in accordance with process C-1 of MIL-P-116. Drying shall be in accordance with the most applicable method of the above specification. These surfaces shall then be coated with type P-9 preservative oil of MIL-P-116, or silicone compound of MIL-S-8660 or MIL-C-21567.

5.2 Packaging, level A. Each igniter shall be packaged in a container conforming to Drawing D4-2-89. The external thread end of the igniter shall be toward the top of the container. The can shall be provided with an interior liner fabricated from paperboard conforming to type I, style 1 of PPP-P-291. The liner shall extend the full diameter and from the bottom to within 1/4 inch of the top of the can. Filler pads fabricated from fiberboard conforming to class domestic, type CF, grade 175 min, C flute, variety SW of PPP-F-320 shall be included, at least one in the bottom and one in the top. Additional filler pads shall be used as required, to restrict free movement of the igniter.

5.3 Packing, level A. Twenty-four igniters, packaged as specified in 5.2, shall be packed in a nailed wood box conforming to class 2, style 4, type 2 (average) load, 100 - 250 lb. load limit of PPP-B-621. All wood material of the box shall be preservative treated by immersing in wood preservative conforming to composition A of TT-W-572 for a minimum period of 1 minute. Prior to use the treated boxes shall meet the preservative dryness requirements specified in MIL-B-2427. The igniters shall be arranged in one layer of four rows of six each. Filler pads fabricated from fiberboard conforming to type CF, class weather-resistant,

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variety SW, grade optional of PPP-F-320 shall be placed inside the box at the sides, ends, top and bottom. Additional pads shall be used as required to make a tight pack. The box shall be closed and strapped in accordance with the applicable requirements of the box specification. Strapping shall conform to 1/2 by 0.020 inches, class B, type I of QQ-S-781.

5.4 Marking. In conjunction with information furnished in the contract or order, marking shall be as specified herein.

5.4.1 Unit package. Each can shall be marked in accordance with MIL-STD-129, color red number 31158. When a decal is used, the color red marking shall be on a color light green number 34558 background. Marking shall be clear and legible after can is closed. All colors shall conform to Fed. Std. No. 595.

5.4.2 Packing (shipping container). Marking of the shipping container shall conform to Drawing D14-5-2830.

5.5 Palletization. When specified in the contract or order, (mandatory for level A) packed igniters shall be palletized and marked in accordance with Drawing D14-5-2829 (see 6.2).

6. NOTES

6.1 Intended use. This item is intended as an ignition element for a fire bomb.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) That the supplier is not required to furnish radiographic equipment.
(c) Palletization when required.

6.3 Precaution. Due to the hazardous nature inherent in filling munitions with chemical agent, suppliers are advised to take all necessary precautions to protect personnel while filling and to prevent fires or explosions.

6.4 Quantity. Igniters filled in accordance with this specification will contain approximately 1-3/8 pounds of WP. This information is furnished as an aid to procurement and not as a criterion for filling.

6.5 Procedure. If igniters are filled under water, the water should be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0 as a corrosion inhibitor.

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6.6 Decontamination. Igniters may be decontaminated by flushing with hot water while vigorously scrubbing with a wire brush.

Custodian:

Army - MU
Navy - AS
Air Force - 70

Review activities:

Navy - AS
Air Force - 70

Preparing activity:

Army - MU(EA)

Project No. 1325-0114

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